

POD Analysis of T

M. F

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Turbulent Pipe Flow

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9-15 Mon 12:39

1. Code Execut

tion and Layout

1.1. L

1. b7.m
2. initSpectral.m
 - reads i
3. $\hookrightarrow$ initEigs.m
 - forms

layout

in binary files, takes eg m-fft

corrMat, finds eigenvalues

1. $\hookrightarrow$ initPod.m

- carries out POD calculations (quadrature, multiplication)
Hellstrom Smits 2017 for Snapshot POD)

2. $\hookrightarrow$ timeReconstructFlow.m

- performs 2d reconstruction + plotSkmr (generates 1d ra

Layout 2

ggf between $\alpha\Phi$) according to Papers (Citriniti George 2000 for Classic POD,

dial graph)

1.3. Importa

`pipe = Pipe();` creates a Pipe Class. As the function

1. `obj.CaseId` - stores properties like Re , rotation number S , experiment
frequently called vectors (`rMat` $r = 1, \dots, 0.5$)
2. `obj.pod` - eigen data, used for calculating POD
3. `obj.solution` - computed POD modes
4. `obj.plt` - plot configuration

ant Switches

ns (above) are called, data is stored in sub-structs:

al flags such as quadrature (simpson/trapezoidal), number of gridpoints,

2. Equations Used

in Code Procedure

2.1. Classic P

The following equations a

$$\int_{r'} \mathbf{S}(k;m;r,r') \, \Phi^{(n)}(k;m;r')$$

$$\mathbf{S}(k;m;r,r') = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}(k$$

$$\alpha^{(n)}(k;m;t) = \int_r \mathbf{u}(k;m;r,t) \Phi$$

OD Equations

re used in the above code.

$$r'\,\mathrm{d}r'=\lambda^{(n)}(k;m)\Phi^{(n)}(k;m;r)$$

$$;m;r,t)\mathbf{u}^*(k;m;r',t)\,\mathrm{d}t$$

$$\mathfrak{L}^{(n)*}(k;m;r)r\,\mathrm{d}r$$

2.2. Classic POD

$$\begin{aligned} &\int_{r'} \underbrace{r^{1/2} S_{i,j}(r,r';m;f) \, r^{1/2}}_{W_{i,j}(r,r';m;f)} \\ &= \underbrace{\lambda^{(n)}(m,f)}_{\hat{\lambda}^{(n)}(m;f)} \underbrace{r^{1/2} \phi_i^{(n)}(r;f)}_{\hat{\phi}_i^{(n)}(r,m;f)} \end{aligned}$$

$$\alpha_n(m;t) = \int_r \mathbf{u}(m;r,t) \eta$$

Equations (Fixed)

$$\underbrace{\phi_j^{*(n)}(r';m;f)r'^{1/2}\mathrm{d}r'}_{\hat{\phi}_j^{\psi(i)}(r';m;f)} \\ \underbrace{\phantom{\phi_j^{*(n)}(r';m;f)r'^{1/2}\mathrm{d}r'}}_{m;f)} \\ f)$$

$$.^{1/2}\Phi_n^*(m;r)dr$$

2.3. Snapshot R

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r, t)$$

$$= \Phi_T^{(n)}(k; m; r) \lambda^{(n)}(k; m$$

$$\mathbf{R}(k; m; t, t') = \int_r \mathbf{u}(k; r$$

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r, t)$$

$$= \Phi_T^{(n)}(k; m; r) \lambda^{(n)}(k; m$$

POD Equations

$$\alpha^{(n)*}(k;m;t)dt$$

)

$$n;r,t)\mathbf{u}^*(k;m;r,t')\,r\,\mathrm{d}r$$

$$\alpha^{(n)*}(k;m;t)\,\mathrm{d}t$$

).

2.4. Reconstruction

The reconstruction of the snapshot q is given by

$$q(\xi, t) - \bar{q}(\xi) \approx \sum_{j=1}^r a_j(t) \varphi_j(\xi)$$

$$q(r, \theta, t; x) = \bar{q}(r, \theta, t; x) +$$

Since the snapshot pod implementation is not error-free, the reconstruction of the snapshot q is given by

$$q(r, \theta, t; x) = \bar{q}(r, \theta, t; x) + (\text{factor})$$

nstruction

tion is given by

$\Rightarrow$

$$\sum_{n=1} \sum_{m=0} \alpha^{(n)}(m;t) \Phi^{(n)}(r;m;x)$$

construction can only be recovered by writing for factor $\gg 0$.

$$r\gamma) \sum_{n=1} \sum_{m=0} \alpha^{(n)}(m;t) \Phi^{(n)}(r;m;x)$$

2.5. Reco

In order to reconstruct in code, `caseId.fluctuation = 'off'`. The

nstruction

this is incorrect. The necessary use of (factor γ) is incorrect

3. Der

To derive the questioned eq

$$\frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r,$$

Substitute $\mathbf{u}_T$ w

$$\frac{1}{\tau} \int_0^\tau \left(\sum_l \Phi_T^{(l)}(k; m; r) \alpha^{(l)} \right.$$

ivation

uation, consider the integral:

$$t)\alpha^{(n)*}(k;m;t)dt.$$

ith its expansion:

$$)\alpha^{(n)*}(k;m;t)\Big)\alpha^{(n)*}(k;m;t)dt.$$

Exchange the order of summation and

$$\sum_l \Phi_{\text{T}}^{(l)}(k; m; r) \left(\frac{1}{\tau} \int_0^\tau \alpha^{(l)} \right)$$

Due to the orthogonality, namely t

$$\langle a^{(n)} \alpha^{(p)} \rangle$$

all terms where $l \neq n$ will vanish, and

$$\Phi_{\text{T}}^{(n)}(k; m; r) \left(\frac{1}{\tau} \int_0^\tau \alpha^{(n)} \right)$$

This derivation assumes the normalization of modes and their orthogonality, form that reveals the spatial structure ($\Phi_{\text{T}}^{(n)}$)

derivation

and integration, and apply orthogonality,

$$\int_0^T \alpha^{(l)}(k; m; t) \alpha^{(n)*}(k; m; t) dt \Bigg).$$

that $\alpha^{(n)}$ and $\alpha^{(p)}$ are uncorrelated

$$= \lambda^{(n)} \delta_{np}$$

and there remains only the $l = n$ term,

$$\int_0^T \alpha^{(n)}(k; m; t) \alpha^{(n)*}(k; m; t) dt \Bigg).$$

along with the eigenvalue relationship to simplify the original integral into a sum of each mode scaled by its significance $(\lambda^{(n)})$.

The cross-correlation tensor $\mathbf{R}$ is defined as $\mathbf{R}(k; m; t, t') = \int_r \mathbf{u}(k; m; r, t) \mathbf{u}(k; m; r, t')$ tensor. The n POD m

$$\lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau \mathbf{u}_T(k; m; r, t) \alpha^{(n)*}(k; r, t) dt$$

erivation

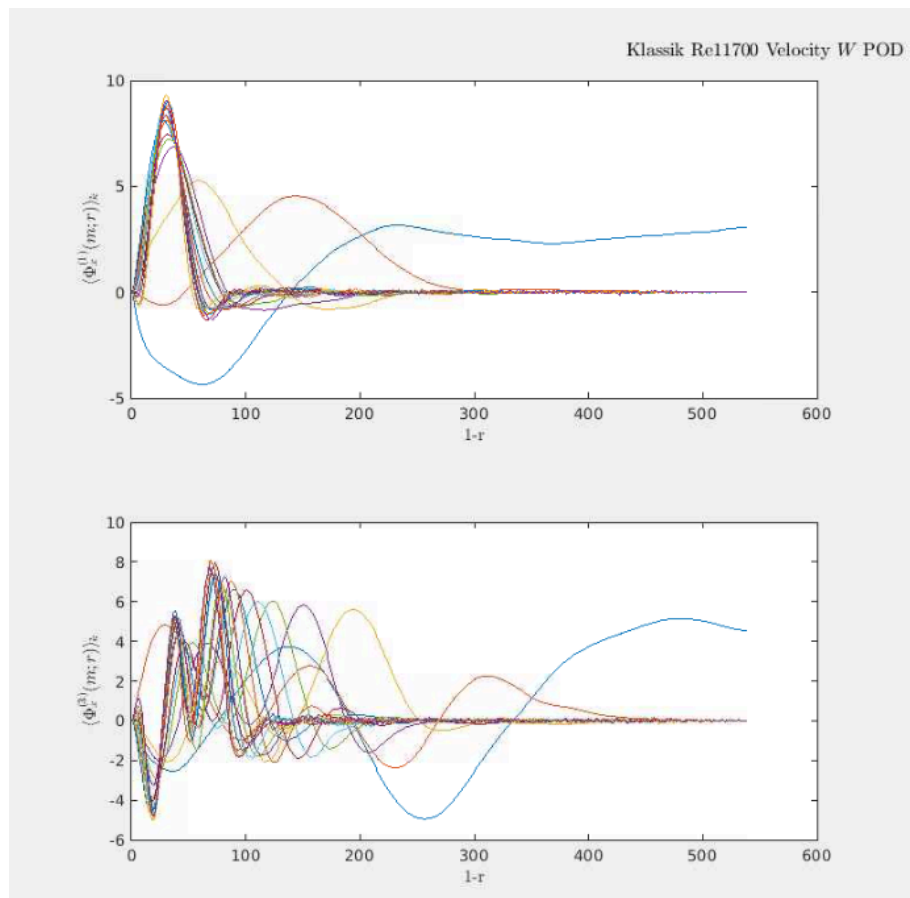
$t)\mathbf{u}^*(k;m;r,t')\,r\,\mathrm{d}r$. This tensor is now transformed from $[3r\times 3r']$ to a
nodes are then constructed as,

$$n;t)\mathrm{d}t=\Phi_{\mathrm{T}}^{(n)}(k;m;r)\lambda^{(n)}(k;m).$$

4. Result Comparison

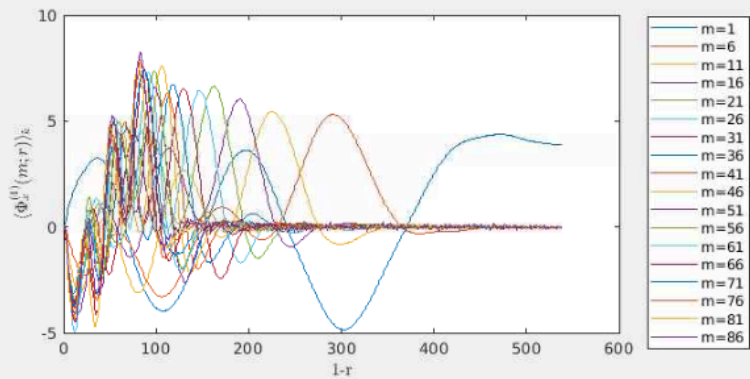
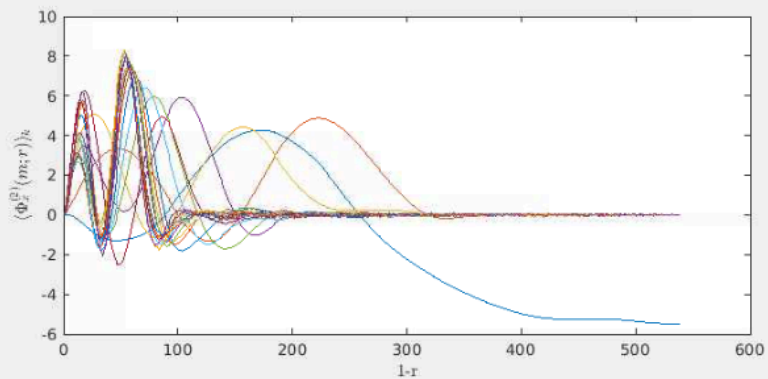
on Classic/Snapshot

4.1. Radi

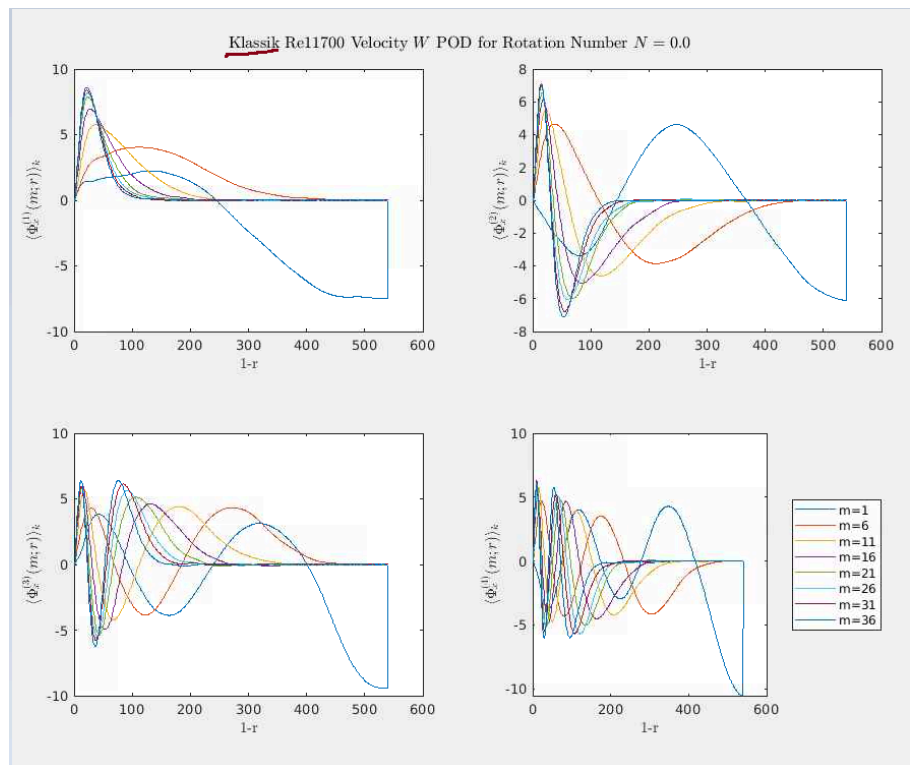


al Classic

for Rotation Number $N = 3.0$

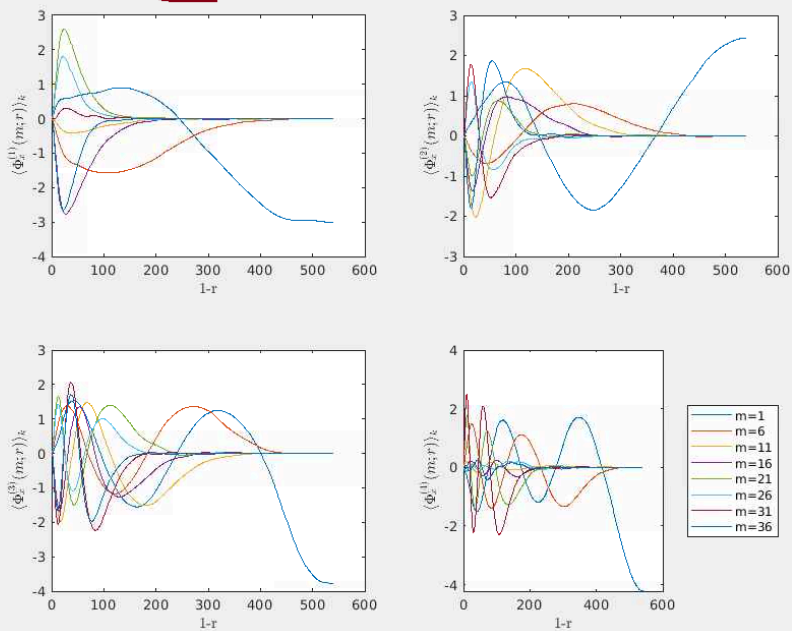


4.2. Snapshot-Cl

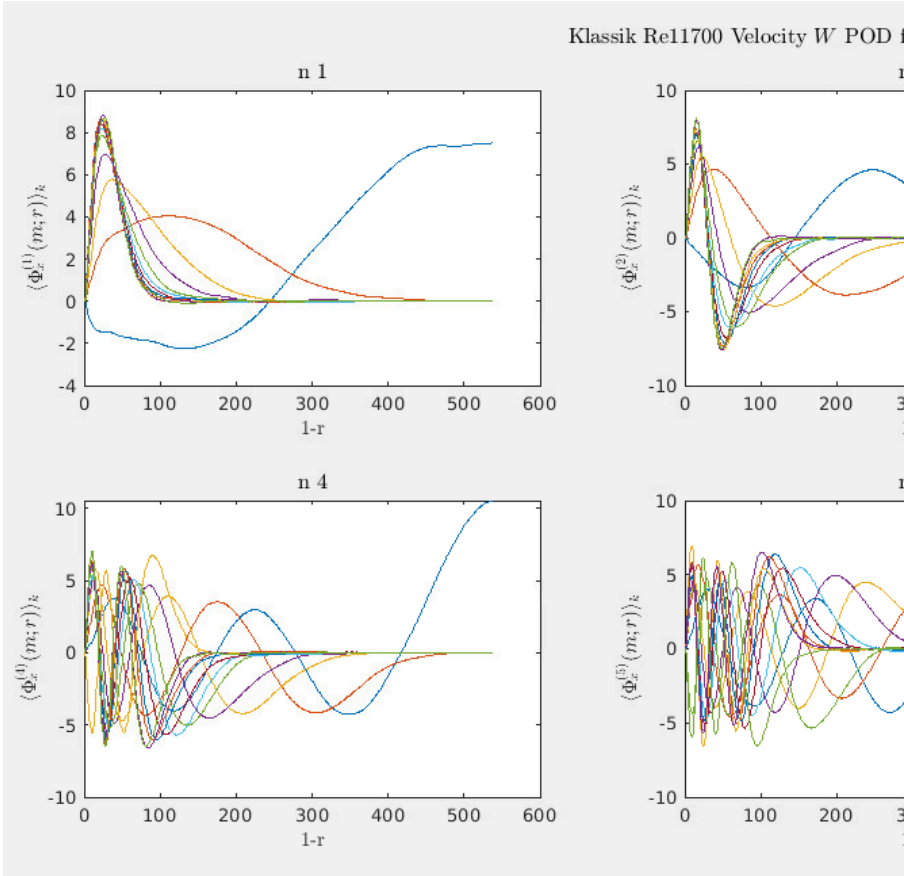


Classic Comparison

Snapshot Rel1700 Velocity W POD for Rotation Number $N = 0.0$



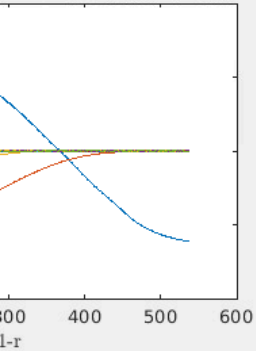
4.3. Klassik



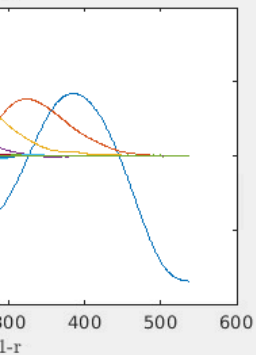
POD S=0.0

For Rotation Number $N = 0.0$

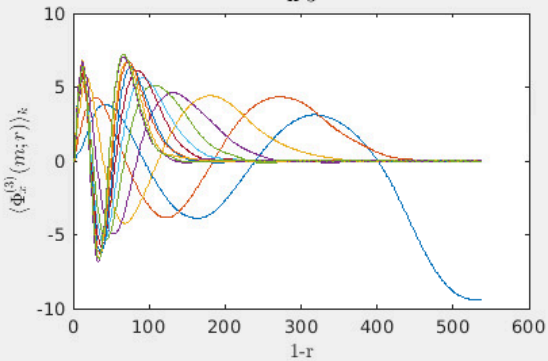
n 2



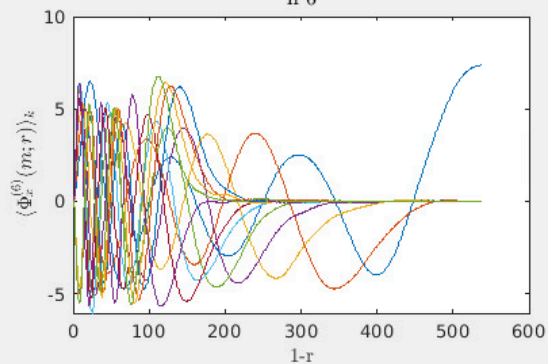
n 5



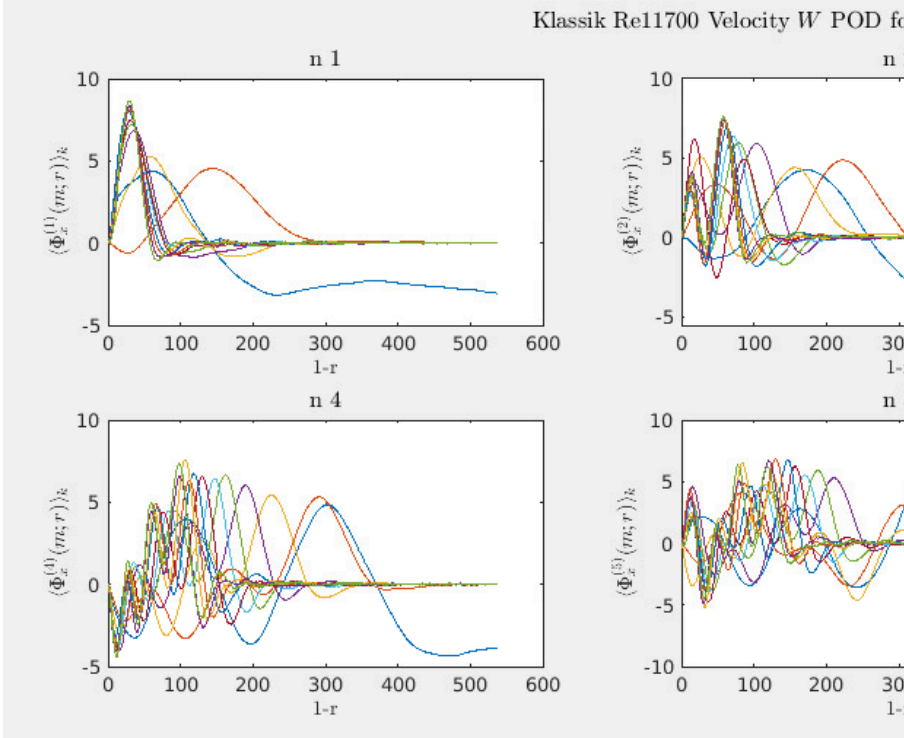
n 3



n 6

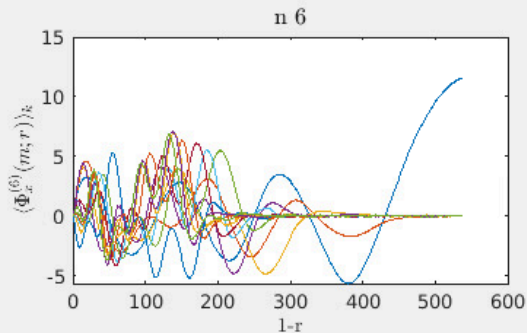
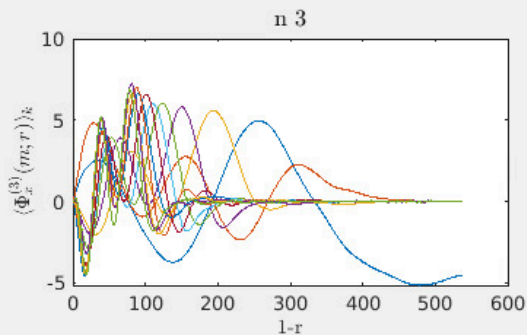
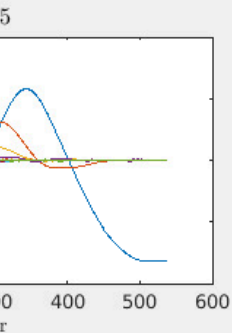
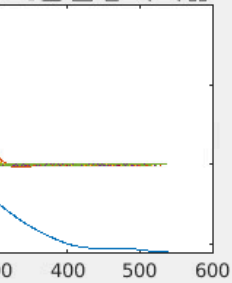


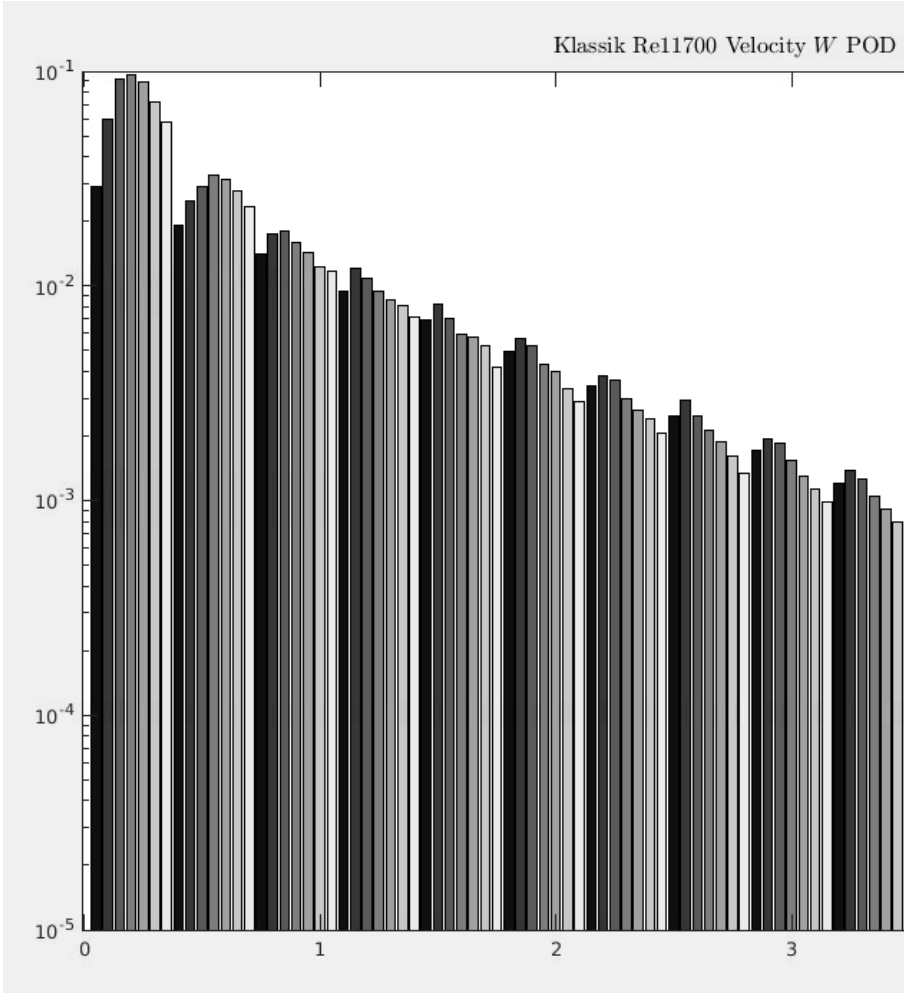
4.4. Klassik



POD S=3.0

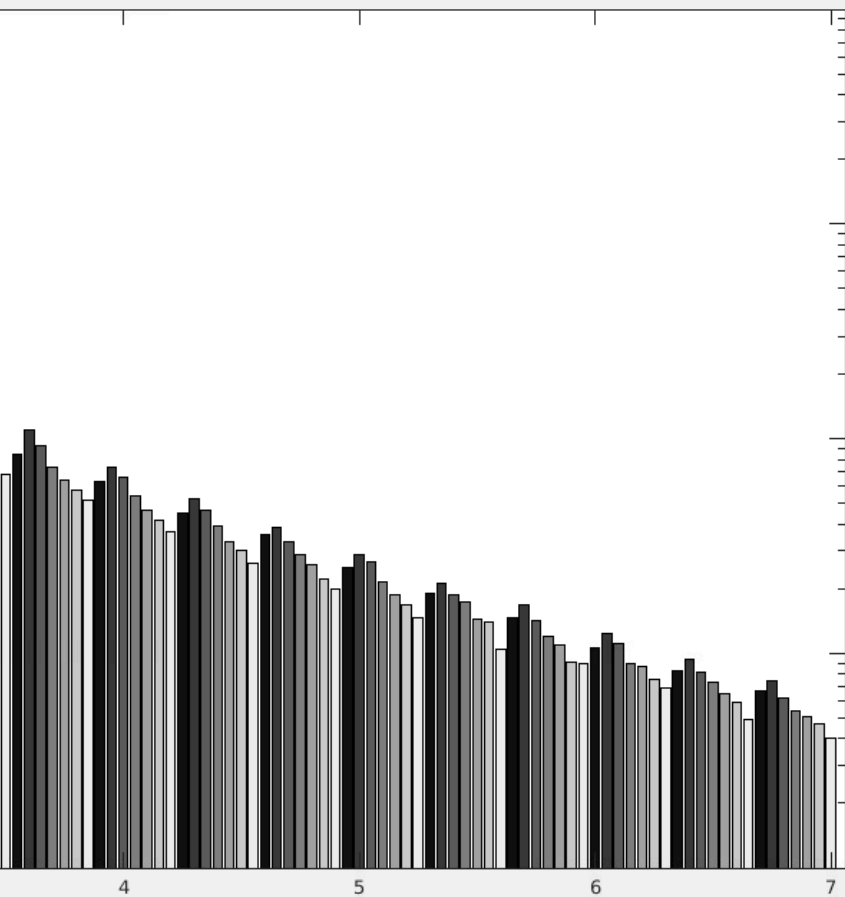
for Rotation Number $N = 3.0$

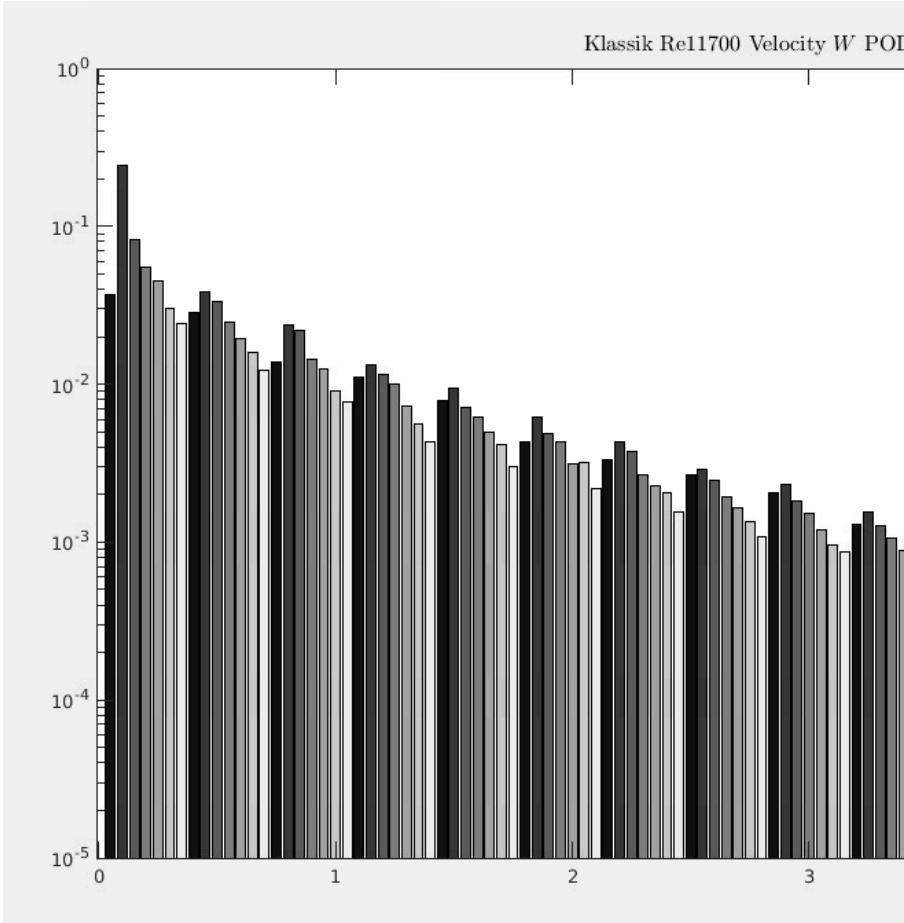




n=0 Classic

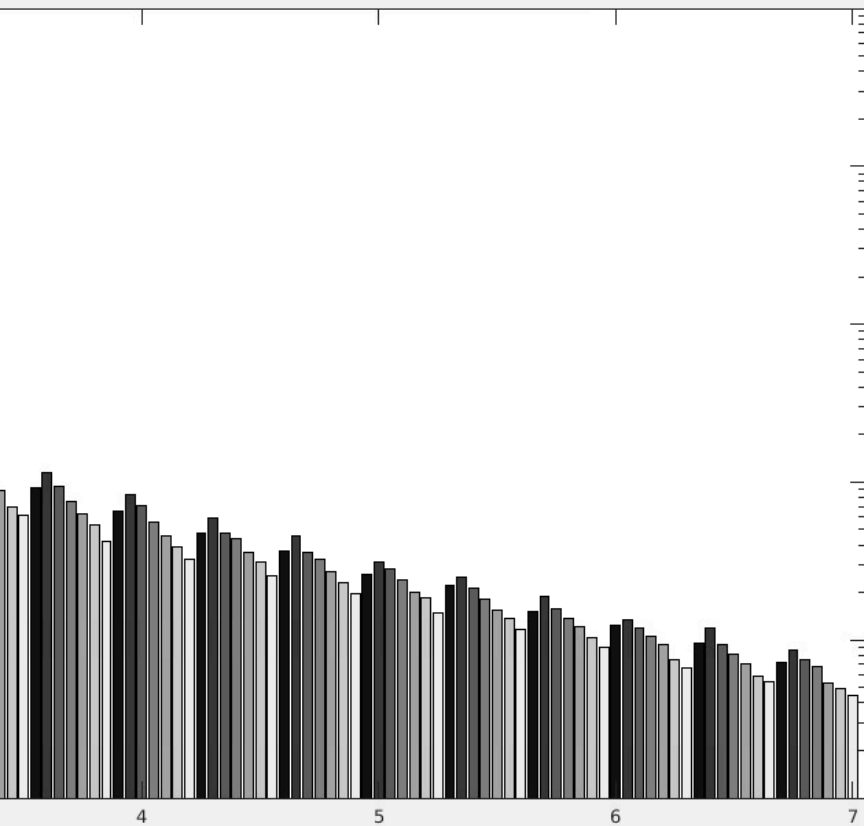
for Rotation Number $N = 0.0$





Classic

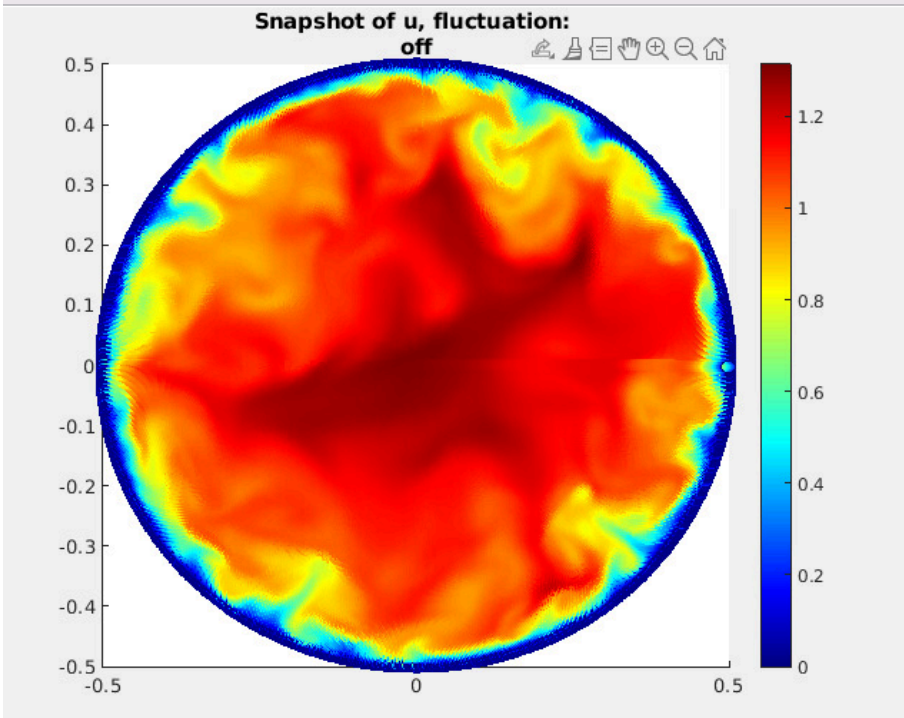
D for Rotation Number $N = 3.0$



analysis

6. Recon

struction



nstruction

REPOD Reconstruction, $t=2$ using $(n,m)=(400,50)$ modes, obf13

